

Credal degrees of freedom for PLN truth values: a layered characterization of strength and confidence

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Abstract

PLN’s two-number truth value $\langle s, c \rangle$ is the compression of a richer object: a credal set of admissible precise previsions over an outcome space. We give a layered formal characterization of where $\langle s, c \rangle$ is *determined* by the spec — the strength is a single real, the confidence is at its information ceiling — and where the underlying credal set has *degrees of freedom* that the two-number summary cannot see.

The characterization rests on a recently-landed Lean dichotomy in `Mettapedia.ProbabilityTheory.ImpreciseProbability.ProjectiveCredal`: the predicates `credalSetDetermines` and `credalSetHasStrictWidth` on a credal set C and a gamble X are the formal locators of, respectively, “ $\langle s, c \rangle$ is fully determined on X ” and “ $\langle s, c \rangle$ has genuine interval width on X .” The collapse theorem `lower_eq_upperEnvelope_of_credalSetDetermines` closes the “determined $\Rightarrow$ precise” direction; the width theorem `lower_upperEnvelope_nontrivial_of_strictWidth` closes the “disagreement $\Rightarrow$ envelope opens” direction.

We lift this dichotomy to the projective credal level (`determinesGlobalGamble / hasStrictGlobalWidth`) and onward to the infinite Markov-logic setting through DLR completions (`dlrQueryDetermined / dlrQueryHasStrictWidth`). Under the Dobrushin uniform small-influence hypothesis, `PaperUniformSmallTotalInfluence` implies `dlrQueryDetermined`, and therefore the MLN query envelope is precise. The low-temperature two-dimensional symmetric-grid theorem now supplies the complementary strict-width crown: `symmetricGridZeroField_originPLNStrictIntervalCrown_of_axisAnchoredContourCode_twentyFour`. On the exchangeable side, the public de Finetti theorem `impreciseDeFinetti_analyticMixingFamily_sharpCompatibilityCrown` separates the unconditional external mixing-family readout from the raw all-gambles crown, which holds exactly at the lower-provision compatibility boundary.

We organize the degrees of freedom into three slots: (A) specification-level freedom in the local credal sets; (B) completion-level freedom in extreme-point choice within a fixed credal set (aleatoric, formally invisible to $\langle s, c \rangle$); (C) phase-transition freedom from non-uniqueness of DLR

completions in infinite MLNs. The endpoint is now a closed Lean package rather than a list of open infinite-side obligations; the remaining freedom is named by the predicates and hypotheses, not hidden in the notation $\langle s, c \rangle$.

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1 Introduction

PLN’s atomic truth-value object is a pair $\langle s, c \rangle$ with $s, c \in [0, 1]$ called *strength* and *confidence*. Standard PLN parametrises s as a probability estimate and c as a function of evidence count, typically $c = N/(N + \kappa)$ for a system-level constant κ and an evidence-count integer N . The pair is what travels through PLN’s rules (revision, deduction, induction, abduction); it is what is read off by query atoms; it is the type signature of every PLN belief.

But $\langle s, c \rangle$ is a two-number summary. The same pair can be produced by very different underlying epistemic objects: a Beta-distribution second-order Bayesian; a Walley-style coherent lower prevision with a specific interval width; an exchangeable de Finetti mixture with a particular concentration. PLN’s compositional rules treat $\langle s, c \rangle$ as the belief object, and the algebra reduces to numerical update formulas; but the question of *what is and is not pinned down by* $\langle s, c \rangle$ is rarely written explicitly in the PLN literature.

This paper makes that explicit. We organise the question as follows. The strength and confidence of a query in PLN are determined by an underlying credal set C of admissible precise previsions. The *degrees of freedom* that $\langle s, c \rangle$ summarises but does not expose live in three slots:

- Specification freedom in the local credal sets that build C ;
- Completion freedom (extreme-point choice) inside a fixed C ;
- Phase-transition freedom from non-uniqueness of global completions when C is the projective limit of a Markov-logic / DLR family.

Each slot has a formally locatable status. Recent landings in the **Mettapedia**.**ProbabilityTheory**.**ImpreciseProbability** and **Mettapedia**.**Logic** libraries (the *credal dichotomy* predicates and the DLR query lifts; see Sec. 4 and 6) make this dichotomy an actual Lean theorem rather than informal prose. Wherever possible we quote the Lean theorem statements verbatim from their source files.

Scope. We treat strength and confidence as the read-off of credal-envelope width and centre. We do *not* address the propagation of $\langle s, c \rangle$ through PLN’s compositional rules (deduction, induction, abduction, revision); that material is in the existing PLN corpus (ξ PLN, ν PLN, the WM-PLN book; see Sec. 11).

The paper’s contribution is the determinacy / freedom characterization at the level of a single truth value, with formal Lean hooks.

Sibling papers. This paper is a sibling of ξ PLN and the *PLN Review* (papers/xiPLN.tex, papers/pln-review.tex). The WM-PLN book papers/wm-pln-book.tex contains an informal section on degrees of freedom versus forcing (§7.4 in the current draft); the present paper offers the formal characterization that the book section calls for.

2 The credal stack: types and definitions

We work in the formal stack of `Mettapededia.ProbabilityTheory.ImpreciseProbability.ProjectiveCredal` (referenced throughout as *ProjectiveCredal.lean*, file path `Mettapededia/ProbabilityTheory/ImpreciseProbability/ProjectiveCredal.lean`). Outcome spaces are arbitrary types Ω ; gambles are bounded real-valued functions on Ω .

2.1 Gambles and previsions

A *gamble* on Ω is a real-valued function on Ω :

```
abbrev Gamble (Omega : Type*) := Omega -> Real
```

A *precise prevision* on Ω is a Walley-coherent linear functional from gambles to reals: the structure `PrecisePrevision` in `ProjectiveCredal.lean` is the standard linear expectation with constant-preservation, positive homogeneity, and additivity built in:

```
structure PrecisePrevision (Omega : Type*) where
  toFun      : Gamble Omega -> Real
  lower_bound : ... -- c <= X (pointwise) implies c <= toFun X
  pos_homog   : ... -- toFun (r . X) = r * toFun X for r >= 0
  add         : ... -- toFun (X + Y) = toFun X + toFun Y
```

A *lower prevision* relaxes additivity to superadditivity, and is the Walley credal-summary object:

```
structure LowerPrevision (Omega : Type*) where
  toFun      : Gamble Omega -> Real
  lower_bound : ... -- c <= X pointwise implies c <= toFun X
  pos_homog   : ... -- toFun (r . X) = r * toFun X for r >= 0
  superadd     : ... -- toFun (X + Y) >= toFun X + toFun Y
```

Coherence (positive homogeneity, superadditivity, and avoidance of sure loss) is the Walley condition for the lower prevision to be the lower envelope of some non-empty set of precise previsions.

2.2 Credal sets and envelopes

A *credal set* is just a set of precise previsions:

```
abbrev CredalPrevisionSet (Omega : Type*) := Set (PrecisePrevision
  Omega)
```

For a credal set C and a gamble X , the *lower* and *upper* envelopes are the standard inf / sup:

```
def lowerEnvelope (C : CredalPrevisionSet Omega) (X : Gamble Omega) :
  Real :=
  sInf ((fun P : PrecisePrevision Omega => P X) '' C)

def upperEnvelope (C : CredalPrevisionSet Omega) (X : Gamble Omega) :
  Real :=
  sSup ((fun P : PrecisePrevision Omega => P X) '' C)
```

The construction `lowerEnvelopePrevision` packages `lowerEnvelope` together with the proofs of constant-preservation, positive homogeneity, and superadditivity, exhibiting a `LowerPrevision` object out of any non-empty bounded-below credal set. The finite case theorem `lowerEnvelopePrevision_isCoherent_of_finite` confirms Walley coherence on finite nonempty Ω .

2.3 PLN $\langle s, c \rangle$ as the credal-envelope summary

The canonical Lean STV type lives in `Mettapedia.Logic.PLNDeduction` at lines 138–139:

```
/-- Simple Truth Value: (strength, confidence) pair -/
structure STV where
  strength    : Real
  confidence  : Real
  strength_nonneg : 0 <= strength
  strength_le_one : strength <= 1
  confidence_nonneg : 0 <= confidence
  confidence_le_one : confidence <= 1
```

Its distributional model is in `Mettapedia.Logic.PLNDistributional`: given a $\text{Beta}(\alpha, \beta)$ distribution, the canonical PLN STV reads

$$s = \frac{\alpha}{\alpha + \beta}, \quad c = f(\alpha + \beta)$$

for an evidence-count map f (typically $f(N) = N/(N + \kappa)$).

The two-number compression from a credal set C and a query gamble X_q (e.g. the indicator $\mathbf{1}_A$ of a query atom A) is the following implicit map:

$$C, X_q \mapsto \begin{cases} s_q = \frac{1}{2}(\text{lowerEnvelope } C X_q + \text{upperEnvelope } C X_q) \\ c_q = g(\text{upperEnvelope } C X_q - \text{lowerEnvelope } C X_q) \end{cases} \quad (1)$$

for a monotone-decreasing $g : [0, 1] \rightarrow [0, 1]$ with $g(0) = 1$ (no width, full confidence) and $g(1) = 0$ (maximum width, no information). The specific choice of

midpoint and g is an interface design decision (PLN traditionally uses midpoint for s and a sigmoidal g derived from the Beta variance); the freedom characterization of this paper is independent of this choice. What matters is that (1) is the read-off, and that the credal set C may have degrees of freedom that the pair (s_q, c_q) cannot expose.

Remark 1. *The summary map (1) loses two kinds of information. First, which gambles have positive credal-width and which do not: (s_q, c_q) summarises a single query but a credal set with non-zero width on some queries may have zero width on others. Second, the shape of C : a credal set on Ω whose extreme points are clustered (high agreement) and one whose extreme points are dispersed (low agreement) can produce the same midpoint and width on a particular query, but compose differently under PLN's inference rules. The width as a function on gambles — which we will study below — is the right object; (s_q, c_q) is its evaluation at one point.*

3 Finite-case determinacy

We first treat the case where the outcome space Ω is finite. Here the Walley theory is complete in the formalization; we summarise the load-bearing results.

3.1 Uniform positivity and avoidance of sure loss

For finite Ω , a pointwise-strictly-positive gamble has a uniform positive lower bound :

```
theorem finite_strictlyPositive_uniformLowerBound
  {Omega : Type*} [Fintype Omega] [Nonempty Omega]
  (X : Gamble Omega) (hX : forall omega, 0 < X omega) :
  exists epsilon : Real, 0 < epsilon and forall omega, epsilon <= X
    omega
```

This is the technical hinge that lets every nonempty finite credal set *avoid sure loss* in the strict Walley sense:

```
theorem lowerEnvelopePrevision_avoidsSureLoss_of_finite
  {Omega : Type*} [Fintype Omega] [Nonempty Omega]
  (C : CredalPrevisionSet Omega) (hC : C.Nonempty)
  (hBdd : forall X : Gamble Omega,
    BddBelow ((fun P : PrecisePrevision Omega => P X) '' C)) :
  (lowerEnvelopePrevision C hC hBdd).avoidsSureLoss
```

and to be *coherent* :

```
theorem lowerEnvelopePrevision_isCoherent_of_finite
  {Omega : Type*} [Fintype Omega] [Nonempty Omega]
  (C : CredalPrevisionSet Omega) (hC : C.Nonempty)
  (hBdd : forall X : Gamble Omega,
    BddBelow ((fun P : PrecisePrevision Omega => P X) '' C)) :
  (lowerEnvelopePrevision C hC hBdd).isCoherent
```

The take-away for PLN: on a finite outcome space, the credal-envelope construction never fails. The map from spec to $\langle s, c \rangle$ is total.

3.2 Singleton collapse

The case where the credal set has exactly one element is the case where PLN's $\langle s, c \rangle$ has no freedom at all; everything is pinned. The theorem is direct:

```
theorem lowerEnvelope_singleton
  {Omega : Type*} (P : PrecisePrevision Omega) (X : Gamble Omega) :
  lowerEnvelope ({P} : CredalPrevisionSet Omega) X = P X
```

and similarly for `upperEnvelope_singleton`. On a singleton credal set, lower = upper on every gamble. PLN-side: every query has c at its information ceiling, and s is the single value PX_q .

4 The credal dichotomy: where strength is determined and where it has width

The contribution of the recent landing in `ProjectiveCredal.lean` is to make the “determined / has width” distinction *predicate-level*, not theorem-level. Two definitions are added (lines 746 and 753), each followed by collapse / non-triviality theorems that pin down the PLN-side consequence.

4.1 The two predicates

Definition 2 (determined credal set on a gamble). *The predicate `credalSetDetermines` :*

```
def credalSetDetermines {Omega : Type*}
  (C : CredalPrevisionSet Omega) (X : Gamble Omega) : Prop :=
  forall P : PrecisePrevision Omega, P in C ->
    forall Q : PrecisePrevision Omega, Q in C -> P X = Q X
```

C determines X when every two admissible precise previsions in C assign the same expectation to X .

Definition 3 (strict width). *The predicate `credalSetHasStrictWidth` :*

```
def credalSetHasStrictWidth {Omega : Type*}
  (C : CredalPrevisionSet Omega) (X : Gamble Omega) : Prop :=
  exists P : PrecisePrevision Omega, P in C and
    exists Q : PrecisePrevision Omega, Q in C and P X < Q X
```

C has strict width on X when there exist two precise previsions in C that strictly disagree on X .

The two predicates are mutually exclusive :

Theorem 4 (mutual exclusion). *`not_credalSetDetermines_of_strictWidth`:*

```

theorem not_credalSetDetermines_of_strictWidth
  {Omega : Type*}
  {C : CredalPrevisionSet Omega} {X : Gamble Omega}
  (hWidth : credalSetHasStrictWidth C X) :
  not (credalSetDetermines C X)

```

The endpoint instances cover the trivial cases. Subsingleton credal sets collapse to determinacy (`credalSetDetermines_of_subsingleton`), and so do singletons (`credalSetDetermines_singleton`):

```

theorem credalSetDetermines_singleton {Omega : Type*}
  (P : PrecisePrevision Omega) (X : Gamble Omega) :
  credalSetDetermines ({P} : CredalPrevisionSet Omega) X

```

4.2 Collapse and width theorems

The PLN-side consequence of `credalSetDetermines` is precisely that strength is pinned: the lower and upper envelopes coincide:

Theorem 5 (envelope collapse from determinacy). *lower_eq_upperEnvelope_of_credalSetDetermines*:

```

theorem lower_eq_upperEnvelope_of_credalSetDetermines
  {Omega : Type*}
  (C : CredalPrevisionSet Omega) (X : Gamble Omega)
  (hC : C.Nonempty)
  (hBddBelow : BddBelow ((fun P => P X) '' C))
  (hBddAbove : BddAbove ((fun P => P X) '' C))
  {P : PrecisePrevision Omega} (hP : P in C)
  (hDet : credalSetDetermines C X) :
  lowerEnvelope C X = upperEnvelope C X

```

This is the formal statement of “ $\langle s, c \rangle$ on X_q is precise” when C determines X_q .

The PLN-side consequence of `credalSetHasStrictWidth` is the opposite: the envelope is genuinely non-trivial :

Theorem 6 (envelope width from strict disagreement). *lower_upperEnvelope_nontrivial_of_strictWidth*:

```

theorem lower_upperEnvelope_nontrivial_of_strictWidth
  {Omega : Type*}
  (C : CredalPrevisionSet Omega) (X : Gamble Omega)
  (hBddBelow : BddBelow ((fun P => P X) '' C))
  (hBddAbove : BddAbove ((fun P => P X) '' C))
  (hWidth : credalSetHasStrictWidth C X) :
  lowerEnvelope C X < upperEnvelope C X

```

This is the formal statement of “ $\langle s, c \rangle$ on X_q is genuinely an interval” when two completions in C strictly disagree on X_q .

4.3 PLN read-off

Putting the dichotomy into the summary map (1): for any credal set C on which both envelope-boundedness conditions hold, and any query gamble X_q , exactly one of the following holds:

predicate	envelope	PLN $\langle s, c \rangle$
<code>credalSetDetermines $C X_q$</code>	$L = U$	precise $s = L X_q$; c at ceiling
<code>credalSetHasStrictWidth $C X_q$</code>	$L < U$	s in interval $[L, U]$; $c < \text{ceiling}$

The Lean dichotomy is the formal locator of which case the PLN truth-value read-off lives in.

5 Projective credal specs: three slots for freedom

Once Ω is large (or infinite), one builds the credal set *compositionally* from finite-window data. The standard packaging is a projective family of local credal sets, indexed by “windows” (finite subsets, finite cylinders) of the global outcome space, glued by restriction maps.

5.1 The projective machinery

The cylinder structure (ProjectiveCredal.lean, the `ProjectiveCylinderSystem` structure) packs the window-to-local projection and the local restriction maps:

```
structure ProjectiveCylinderSystem (Window Global : Type*) [LE Window]
  where
    Local      : Window -> Type*
    project    : forall i : Window, Global -> Local i
    restrict   : forall {i j : Window}, i <= j -> Local j ->
      Local i
    project_restrict : ... -- compatibility on the diagram
```

The full spec layers a local credal choice on top:

```
structure ProjectiveLocalCredalSpec (Window Global : Type*) [LE Window] where
  cylinders : ProjectiveCylinderSystem Window Global
  localCredal : forall i : Window, CredalPrevisionSet (cylinders.
    Local i)
```

The *global credal set* associated to a spec is the `projectiveLimitCredalSet` (the set of P in `PrecisePrevision Global` that marginalise into `localCredal i` at every window i).

5.2 Three identifying cases and the dichotomy at the global level

The codebase ships two endpoints and one general construction:

- **Singleton :**

```
def singletonIdentityProjectiveSpec {Omega : Type*}
  (P : PrecisePrevision Omega) : ProjectiveLocalCredalSpec
  PUnit Omega
```

Every window’s local credal is the supplied singleton $\{P\}$. The projective limit is $\{P\}$ (`singletonIdentityProjectiveSpec_projectiveLimitCredalSet`). PLN $\langle s, c \rangle$ on every gamble is precise; this is the “no freedom” case.

- **Unrestricted :**

```
def finiteUnrestrictedProjectiveSpec (Omega : Type*) :
  ProjectiveLocalCredalSpec PUnit Omega
```

Every window’s local credal is `Set.univ`. The projective limit contains every precise prevision on Ω . PLN $\langle s, c \rangle$ on most gambles has `credalSetHasStrictWidth`; this is the “maximum freedom” case. The non-emptiness of the projective limit is witnessed by `finiteUnrestricted_finiteEvaluationCompact_hasCompatibleCompletion` via a compact carrier in the finite-evaluation topology.

- **Genuine intermediate:** any spec whose local credals are non-trivial proper subsets of `Set.univ`. Whether `credalSetDetermines` or `credalSetHasStrictWidth` holds on a particular gamble depends on the spec and the gamble.

The dichotomy lifts to the projective level. `ProjectiveCredal.lean` defines `determinesGlobalGamble` and `hasStrictGlobalWidth` as the global analogues of `credalSetDetermines` and `credalSetHasStrictWidth`, with the mutual exclusion theorem `not_determinesGlobalGamble_of_strictWidth`. The natural-extension lifting theorems connect the projective predicates to the lower / upper envelopes on the projective limit.

5.3 The three formal slots, structurally

We can now name the three slots in which PLN credal freedom lives. Each is formally locatable through one of the predicates above.

- (A) **Specification-level freedom.** The choice of `localCredal` i at each window. This is the modeller’s freedom: evidence width, prior choices, and what classes of measure are taken to be admissible at each local scale.

- (B) **Completion-level freedom.** Given a fixed credal set C , the choice of $P \in C$ that actually governs the world. This freedom is purely aleatoric: it is invisible to $\langle s, c \rangle$ because $\langle s, c \rangle$ is a property of C , not of any specific $P \in C$.
- (C) **Phase-transition freedom.** For globally-glued projective specs, the projective limit may itself have multiple admissible elements that correspond to different *global* solutions of the same local problem. This is the slot Markov-logic / DLR exemplifies (next section).

Slots (A) and (C) are the only ones $\langle s, c \rangle$ can see. Slot (B) is formally invisible at the truth-value level.

6 Infinite Markov logic: Dobrushin collapse and DLR width

For the infinite case — countably-many ground atoms, Markov-logic network with weights, density specified locally — the projective machinery specialises into the Dobrushin–Lanford–Ruelle framework. The file `Mettapedia.Logic.MarkovLogicInfiniteCredalBridge` (`Mettapedia/Logic/MarkovLogicInfiniteCredalBridge.lean`) packages this.

6.1 Classical MLN spec and DLR completions

A `ClassicalInfiniteGroundMLNSpec` packages a countable atom set, a clause-id set, the clauses, their log-weights, and the local finite-volume region structure. A *DLR completion* of such a spec is a probability measure on the infinite world satisfying the DLR consistency condition at every finite volume:

```
abbrev DLRCompletion (M : ClassicalInfiniteGroundMLNSpec Atom
  ClauseId) :=
  { mu : ProbabilityMeasure (InfiniteWorld Atom)
    // FixedRegionCylinderDLR M.toStrictlyPositiveInfiniteGroundMLNSpec
    (mu : Measure (InfiniteWorld Atom)) }
```

The set of DLR completions of M is the “credal set” for the infinite spec, and the projective-credal lift packages it as `DLRProjectiveCredalSpecialization` in `Mettapedia/Logic/MarkovLogicInfiniteCredalBridge.lean`.

A *query* is a finite-volume event; its *probability* under a DLR completion is its measure:

```
noncomputable def dlrCompletionQueryProb
  (M : ClassicalInfiniteGroundMLNSpec Atom ClauseId)
  (q : ConstraintQuery Atom) (mu : DLRCompletion M) : Real :=
  ENNReal.toReal
  ((infiniteMLNMassSemantics M mu.1 mu.2).queryProb q)
```

6.2 DLR-level dichotomy

The credal dichotomy lifts directly to the DLR setting:

```
def dlrQueryDetermined
  (M : ClassicalInfiniteGroundMLNSpec Atom ClauseId)
  (q : ConstraintQuery Atom) : Prop :=
  forall mu nu : DLRCompletion M,
    dlrCompletionQueryProb M q mu = dlrCompletionQueryProb M q nu

def dlrQueryHasStrictWidth
  (M : ClassicalInfiniteGroundMLNSpec Atom ClauseId)
  (q : ConstraintQuery Atom) : Prop :=
  exists mu : DLRCompletion M, exists nu : DLRCompletion M,
    dlrCompletionQueryProb M q mu < dlrCompletionQueryProb M q nu
```

`dlrQueryDetermined` says all DLR completions agree on the query; `dlrQueryHasStrictWidth` says two of them strictly disagree. The mutual exclusion theorem `not_dlrQueryDetermined_of_strictWidth`:

```
theorem not_dlrQueryDetermined_of_strictWidth
  (M : ClassicalInfiniteGroundMLNSpec Atom ClauseId)
  (q : ConstraintQuery Atom)
  (hWidth : dlrQueryHasStrictWidth M q) :
  not (dlrQueryDetermined M q)
```

6.3 Dobrushin uniqueness as the collapse-direction hypothesis

The classical Dobrushin uniform-influence condition is packaged as the predicate `PaperUniformSmallTotalInfluence` on the MLN spec in `Mettapedia.Logic.MarkovLogicInfiniteUniqueness`. The collapse direction of the DLR-level dichotomy is the theorem `dlrQueryDetermined_of_uniform`:

```
theorem dlrQueryDetermined_of_uniform
  (M : ClassicalInfiniteGroundMLNSpec Atom ClauseId)
  (hM : M.PaperUniformSmallTotalInfluence)
  (q : ConstraintQuery Atom) :
  dlrQueryDetermined M q
```

Under the Dobrushin uniform small-influence hypothesis, every finite query is DLR-determined. Combining with the dichotomy gives the precision of the MLN query envelope:

```
theorem infiniteMLN_queryEnvelope_precise_of_uniform
  (M : ClassicalInfiniteGroundMLNSpec Atom ClauseId)
  [Nonempty (DLRCompletion M)]
  (hM : M.PaperUniformSmallTotalInfluence)
  (q : ConstraintQuery Atom) :
  infiniteMLNLowerQueryEnvelope M q
  = infiniteMLNUpperQueryEnvelope M q
```

Under the same hypothesis, and assuming the DLR completion family is inhabited, the PLN $\langle s, c \rangle$ for every finite query is precise.

6.4 Strict width as the phase-transition canary

The opposite direction is the theorem `infiniteMLN_queryEnvelope_nontrivial_of_disagreement` and the lift through `infiniteMLN_queryEnvelope_nontrivial_of_strictWidth`:

```
theorem infiniteMLN_queryEnvelope_nontrivial_of_strictWidth
  (M : ClassicalInfiniteGroundMLNSpec Atom ClauseId)
  (q : ConstraintQuery Atom)
  (hBddBelow : BddBelow (Set.range (dlrCompletionQueryProb M q)))
  (hBddAbove : BddAbove (Set.range (dlrCompletionQueryProb M q)))
  (hWidth : dlrQueryHasStrictWidth M q) :
  infiniteMLNLowerQueryEnvelope M q
    < infiniteMLNUpperQueryEnvelope M q
```

If two DLR completions disagree on a finite query, the PLN $\langle s, c \rangle$ for that query is genuinely an interval.

6.5 Dobrushin collapse and strict-width witnesses

Combining the two proved directions gives a one-sided collapse theorem and a separate strict-width witness theorem. The Dobrushin condition `PaperUniformSmallTotalInfluence` is a formally sufficient hypothesis for query determinacy and precise PLN/Walley envelopes. Separately, if two DLR completions are exhibited that disagree on a query, then that query has strict width. The Lean development does not prove that failure of the Dobrushin hypothesis entails non-uniqueness or strict width.

7 Exchangeable de Finetti families: the F2 compatibility boundary

The second infinite lane is the exchangeable IID/de Finetti lane. Here the formal point is not merely that an external mixture representation exists: it does, unconditionally, for any nonempty credal family of Bernoulli mixtures. The stronger all-gambles process-law crown has a sharper boundary.

The public theorem `impreciseDeFinetti_analyticMixingFamily_sharpCompatibilityCrown` states:

```
theorem
  impreciseDeFinetti_analyticMixingFamily_sharpCompatibilityCrown
  (C : Set BernoulliMixture) (hC : C.Nonempty) :
  ImpreciseDeFinettiCanonicalExternalMixingFamily C hC /\
    (ImpreciseDeFinettiAnalyticMixingFamilyProcessLawCrown C hC <->
      exists L : LowerPrevision (Nat -> Bool),
        (bernoulliMixturePrefixProcessLowerSpec C ... hC).
          respectsLocalLower L)
```

Thus the canonical sigma-additive external readout is always present, while the raw all-gambles analytic crown is pinned exactly by existence of a respecting

lower prevision for the analytic prefix lower envelope. This is the Walley natural-extension boundary in the exchangeable process setting.

A useful negative theorem prevents over-reading the result. The theorem `exists_impreciseDeFinetti_analyticMixingFamilyProcessLawCrown_not_pointwiseZeroInterior` exhibits a nonempty mixing family whose raw all-gambles crown exists even though not every member has zero interior mixing mass. Equivalently, `not_forall_impreciseDeFinetti_analyticMixingFamilyProcessLawCrown_imp_pointwiseZeroInterior_closed` closes the refutation of the too-strong pointwise-zero necessity claim. Pointwise zero-interior is a sufficient concrete route; it is not the right family-level characterization.

8 Worked examples already in the formalization

The codebase now ships both finite toy witnesses and a concrete infinite DLR/PLN contrast crown, plus the low-temperature symmetric-grid strict interval crown. The finite two-spin examples remain useful sanity witnesses for strict credal width.

8.1 Two-spin magnet

In `ProjectiveCredal.lean` around lines 1064–1101, the type `TwoSpin`, defined as `Bool × Bool`, carries the two-spin magnet credal set

```
def twoSpinMagnetCredalSet : CredalPrevisionSet TwoSpin :=
  { PrecisePrevision.dirac twoSpinAllDown,
    PrecisePrevision.dirac twoSpinAllUp }
```

The credal set contains two Dirac measures: all spins down, and all spins up. The first-spin-up gamble

```
def twoSpinFirstUpGamble : Gamble TwoSpin :=
  fun s => if s.fst then 1 else 0
```

is rewarded by the all-up Dirac and unrewarded by the all-down Dirac; the disagreement gives the strict width witness. The packaged theorem `twoSpinMagnetEnvelope_nontrivial` closes:

```
theorem twoSpinMagnetEnvelope_nontrivial :
  lowerEnvelope twoSpinMagnetCredalSet twoSpinFirstUpGamble
  < upperEnvelope twoSpinMagnetCredalSet twoSpinFirstUpGamble
```

PLN-side: on this two-Dirac credal, the magnetization query has lower envelope 0, upper envelope 1, and confidence at its floor.

8.2 Zero-temperature aligned magnet

An incremental sibling `twoSpinZeroTemperatureMagnetEnvelope_nontrivial` adds the zero-temperature Gibbs constraint (spins must be aligned), and shows

the envelope is still strictly non-trivial: any measure supported on the aligned-spin pair $\{(\downarrow, \downarrow), (\uparrow, \uparrow)\}$ still admits both extremes. The structural constraint by itself does not close down credal width; the Dobrushin condition has to be checked *quantitatively* on the local-influence row-sums.

8.3 Concrete infinite crown: reinforced line versus zero-weight grid

The theorem `reinforcedLineGeometric_strictWidth_and_zeroWeightGrid_collapse` packages the first concrete DLR/PLN contrast. The strict-width side uses `reinforcedLineGeometric_originSpinUp_plnStrictInterval`: a geometrically reinforced half-line has positive PLN strict-width at the origin-spin query. The collapse side uses `gridZeroField_originSpinUp_queryEnvelope_precise_of_dobrushin`: the zero-weight zero-field grid lies in the Dobrushin collapse regime for the analogous origin-spin query. This remains a useful canary for the collapse direction.

8.4 Low-temperature symmetric-grid strict interval crown

The full two-dimensional low-temperature strict-width side is supplied by the symmetric-grid Peierls development in `Mettapedia.Logic.MarkovLogicInfiniteSymmetricGridExample`. The public theorem `symmetricGridZeroField_originPLNStrictIntervalCrown_of_axisAnchoredContourCode_twentyFour` states:

<pre>theorem symmetricGridZeroField_originPLNStrictIntervalCrown_of_axisAnchoredContourCode_twentyFour : SymmetricGridZeroFieldOriginPLNStrictIntervalCrown (24 : Real)</pre>

The crown packages positive scalar query-envelope width, width-complement below one, and positive binary query-outcome credal width for the origin-spin query. The proof path is the classical Peierls route specialized to the finite box: axis-anchored contour codes, a finite-box $\mathbb{F}_2$ separator/potential argument, a geometric contour count, and the resulting plus/minus boundary separation. The theorem is independent of the refuted Markov de Finetti branch and does not rely on the older conditional `CutWalkCode` adapters.

9 Three kinds of degree of freedom: the synthesis

We now name the three formally-locatable freedom slots, each with the predicate that witnesses it.

(A) Specification-level: localCredal choice. A wider local credal at any cylinder window produces a wider projective limit. The PLN-side question “does my $\langle s, c \rangle$ on query X_q have genuine width?” reduces to “does

`hasStrictGlobalWidth spec` on X_q hold?”. The modeller controls this slot by varying `localCredal i`. The slot is finite or infinite depending on whether Ω is.

(B) Completion-level: extreme-point choice. Given a credal set C , the actual world chooses one $P \in C$ (the “true” precise prevision). This freedom is aleatoric. It is formally invisible to $\langle s, c \rangle$ because $\langle s, c \rangle$ is a property of C , not of any specific P . PLN inference rules do not see it; only the verification of a query against actual data sees it.

(C) Phase-transition: DLR non-uniqueness. In infinite Markov-logic specs, the projective limit is the set of DLR completions, which may itself be non-singleton. The slot is opened exactly when `dlrQueryHasStrictWidth` on (M, q) holds for the query q of interest, and closed when `dlrQueryDetermined` on (M, q) holds. The Dobrushin condition `PaperUniformSmallTotalInfluence` is a sufficient condition for the closed case; the low-temperature symmetric-grid Peierls theorem is the complementary proved strict-width case.

Observation 7. *Slot (A) is the freedom the PLN modeller intends to give. Slot (B) is the freedom the world has within the modeller’s spec. Slot (C) is the freedom that the infinite structure of the spec itself produces, without any modeller intent and without any world-side aleatoric residual.*

The three slots are independent. A modeller can pick (A) tight, the world can pick (B) freely, and the spec can still have (C) freedom from a phase transition. The PLN $\langle s, c \rangle$ sees (A) and (C); it does not see (B).

10 Closed endpoint and remaining polish

The current Lean endpoint is the profile `confidenceCharacterizationEndpointProfile`, exported through `Mettapedia.Logic.PLNCanonicalAPI`. It packages the finite forcing results, the typed-STV canaries showing residual degrees of freedom, the Walley width-complement forcing theorem for the PLN odds coordinate, the finite exact-predictive ITV collapse, the DLR/Dobrushin collapse lane, the F2 exchangeable mixing-family compatibility boundary, and the F4 low-temperature symmetric-grid strict-interval crown.

The characterization should therefore be read as closed relative to its explicit semantic gates:

- **Finite forcing.** Once evidence counts are read through a Walley width-complement compatible evidence-weight coordinate, the odds form of PLN confidence is forced by `walley_width_complement_forces_pln_odds`. The free parameter is the evidence-scale constant, not the shape of the coordinate.
- **Finite exact collapse.** Singleton and exact posterior carriers have lower envelope equal to upper envelope. Their ITV readout has width zero, credibility one, and midpoint equal to the single predictive prevision.

- **F2, exchangeable IID/de Finetti.** `impreciseDeFinetti_analyticMixingFamily_sharpCompatibilityCrown` proves that the external sigma-additive mixing-family readout is unconditional, while the raw all-gambles process-law crown holds exactly at the lower-prevision compatibility boundary. The theorem `not_forall_impreciseDeFinetti_analyticMixingFamilyProcessLawCrown_imp_pointwiseZeroInterior_closed` prevents the common but false strengthening to pointwise zero-interior necessity.
- **F3, Dobrushin collapse.** `infiniteMLN_paperUniformSmallTotalInfluence_implies_uniqueMeasure` gives the Markov-logic/DLR collapse direction: uniform small influence forces a unique infinite measure, hence precise finite-query envelopes.
- **F4, low-temperature Ising width.** `symmetricGridZeroField_originPLNStrictIntervalCrown_of_axisAnchoredContourCode_twentyFour` gives the 2D symmetric-grid strict-interval crown at low temperature, by the finite-box Peierls/ $\mathbb{F}_2$ contour route.

What remains is integration polish rather than a missing crown theorem: the book chapter should import the closed endpoint, and future work can broaden the menu of examples and sharpen constants. A runnable companion now accompanies this paper at `papers/pln-confidence-examples/`: it pairs each degree of freedom with a small PeTTa/CeTTa canary and its Lean oracle (see Appendix B), with a one-command harness `run_all.sh`. Those refinements do not change the status of the confidence-formula characterization itself.

11 Related work

Walley. The credal-set lower-prevision framework is Walley (1991). Our `LowerPrevision` structure is the Walley structure with constant-preservation, positive homogeneity, and superadditivity, and our `lowerEnvelopePrevision_isCoherent_of_finite` is the classical Walley coherence theorem for finite envelopes.

de Finetti. The de Finetti exchangeability theorem connects exchangeable distributions on infinite sequences to mixtures over IID distributions. The codebase now has the finite-prefix imprecise de Finetti payoff theorems, the canonical external sigma-additive mixing-family object, and the raw all-gambles process-law compatibility boundary. The important caution is negative: family-level crown existence does not force pointwise zero-interior mass for every mixture in the family.

Dobrushin–Lanford–Ruelle. The DLR framework characterises infinite Gibbs measures by their finite-volume conditional distributions. The Dobrushin uniqueness condition (uniform smallness of total influence) implies a

unique DLR completion. Singla and Domingos have a Markov-logic specific uniformity-of-influence framing. Our `PaperUniformSmallTotalInfluence` is the codified hypothesis.

PLN literature. The two-component truth value $\langle s, c \rangle$ is canonical in the PLN literature (Goertzel et al. *Probabilistic Logic Networks*). The sibling papers ξ PLN (`papers/xiPLN.tex`) and the PLN Review (`papers/pln-review.tex`) treat the evidence-algebra and formal-rule-coherence axes. The WM-PLN book (`papers/wm-pln-book.tex`) has the informal predecessor of the degrees-of-freedom characterization in its §7.4 (“Degrees of freedom versus forcing”).

Markov logic networks. Richardson and Domingos (Machine Learning, 2006) introduced MLNs as finite log-linear models on first-order logic. The infinite specialisation in `Mettapedia.Logic.MarkovLogicInfinite*` is cited inline above: Dobrushin uniqueness supplies the collapse side, and the symmetric-grid Peierls theorem supplies the low-temperature strict-width side.

Imprecise probability and second-order Bayesianity. The PLN tradition of treating c as “confidence-in-strength” has a natural Bayesian second-order reading (a Beta distribution over the true probability), which the file `Mettapedia.Logic.PLNDistributional` formalises. The credal reading we develop here is the Walley-side complement; in the finite case the two readings overlap on the singleton case and diverge on the strict-width case.

12 Conclusion

PLN truth values $\langle s, c \rangle$ are summary statistics of a credal set, and the question of where their information lives reduces, formally, to a dichotomy on the underlying credal set. The codebase landings in `Mettapedia.ProbabilityTheory.ImpreciseProbability.ProjectiveCredal` (the `credalSetDetermines / credalSetHasStrictWidth` predicates and their collapse / width theorems), `Mettapedia.Logic.DeFinettiProjectiveCredalBridge` (the F2 compatibility boundary), and `Mettapedia.Logic.MarkovLogicInfiniteSymmetricGridExample` (the F4 strict-width crown) make this dichotomy a checkable Lean endpoint rather than informal commentary.

The three freedom slots — specification, completion, phase-transition — partition the question along orthogonal axes. (A) is the modeller’s freedom. (B) is the world’s freedom. (C) is the structural freedom the spec produces by itself. PLN $\langle s, c \rangle$ sees (A) and (C); it does not see (B). Where it sees (C), Dobrushin uniqueness pins the high-temperature collapse side and the Peierls theorem pins the low-temperature strict-width side.

The infinite half is no longer a list of named crown gaps. The exchangeable F2 lane is pinned by lower-provision compatibility rather than pointwise zero-interior mass; the DLR F3 lane is pinned on the collapse side by Dobrushin

uniqueness; and the Ising F4 lane is pinned on the width side by the low-temperature symmetric-grid strict-interval crown. What remains is the ordinary work of presentation, API consolidation, and additional examples, not a missing theorem in the confidence-formula characterization.

A Degrees-of-freedom zoo

This appendix lists the degrees of freedom that should remain visible in PLN truth-value practice. Defaults are useful; unnecessary fixation is not. Each entry names the freedom, the formal gate that can collapse it, and why a real system may want to keep the knob available.

Evidence scale / prior strength. In the evidence-count view, confidence is commonly $c = N/(N + \kappa)$, where N is total retained evidence and $\kappa > 0$ is the effective prior or context size. The Walley width-complement bridge forces the odds form once the scale is chosen, but it does not choose the scale. A curated clinical fact, a noisy high-throughput association, and an indirect pathway-mediated hypothesis should not usually accumulate confidence at the same rate. Positive example: a hand-curated contraindication can use small κ . Negative example: a single weak omics hit should not receive the same confidence merely because it has the same positive/negative ratio.

Strength projection / prior context. The same positive and negative counts can be read by maximum likelihood, by a uniform Beta posterior mean, by Jeffreys’ prior, or by another explicit context. The formal tower records this as a projection choice rather than pretending that “strength” is context-free. Positive example: sparse medical counts may need shrinkage toward a population baseline. Negative example: a mature benchmark with thousands of observations can safely use a near-MLE projection.

Evidence total versus evidence ratio. Strength reads the positive/negative ratio; confidence reads the total amount of evidence. Thus 1/1 and 100/100 have the same strength but should not have the same confidence. Conversely, 1/1 and 2/0 can have the same total evidence and different strengths. This is the basic reason the two coordinates are separate.

Confidence chart. Count reconstruction only requires an invertible chart on total evidence; it does not by itself force the PLN odds chart. The width-complement bridge is the extra semantic law that narrows the chart to the PLN/NARS odds coordinate. Positive example: user-facing risk displays may want a compressed visual confidence scale. Negative example: revision must operate in evidence-weight space, not in an arbitrary display chart.

Width-complement convention. For unit gambles, the convention $c = 1 - (U - L)$ makes confidence the complement of interval width. For a general bounded gamble with range $[a, b]$, the normalized version is $c = 1 - (U - L)/(b - a)$. This convention is natural and minimal, but it is still a convention. Positive example: binary prediction has a canonical unit scale. Negative example: dollar-valued utilities must be range-normalized before width-complement confidence is meaningful.

Interval point selector. A nontrivial interval $[L, U]$ does not force a single displayed point: lower, midpoint, upper, or a risk-adjusted selector can all be reasonable. The paper uses midpoint when it wants a neutral PLN strength display. Positive example: safety-critical planning may use the lower envelope. Negative example: exploratory ranking may use the midpoint with the interval width shown separately.

Credal shape beyond one interval. A single query interval loses the shape of the credal set. Two credal sets can agree on L and U for one gamble but differ on other gambles, on correlations, or under composition. Positive example: two drug hypotheses can share the same marginal response interval but differ in adverse-event dependence. Negative example: reporting only an STV for each marginal can hide incompatible joint structure.

Backend for uncertainty. Bayesian credible intervals, Walley IDM predictive intervals, finite lower envelopes, projective credal envelopes, and DLR completion envelopes all produce interval-like readouts but carry different provenance. Positive example: a posterior credible interval is right for a parametric lab assay. Negative example: a robust safety envelope should not be collapsed into a single Bayesian posterior merely for convenience.

Credible level / tolerance. Bayesian credible intervals carry a chosen level, e.g. 90%, 95%, or 99%. This is not the same knob as κ : κ governs evidence accumulation; credible level governs the interval requested from a posterior. Positive example: exploratory dashboards can show 90% intervals. Negative example: clinical decision support may require 99% or a one-sided lower bound.

Categorical query choice. In multinomial or categorical evidence, total concentration can be shared while each queried category has its own lower, upper, width, and strength. Positive example: a phenotype classifier may have high total evidence but low confidence for a rare subclass. Negative example: forcing one binary query can erase category-specific uncertainty.

Feature/evidence discounting. Evidence can be reweighted by source quality, feature informativeness, IDF-like rarity, route length, recency, or dependence corrections before it enters the count tower. Positive example: independent curated replications should count more than near-duplicate text-mined hits. Negative example: treating correlated pathway hits as independent overstates confidence.

Local credal specification. In projective credal models, the modeller chooses local credal sets at finite windows. This is the intended imprecision knob: how much local ambiguity the model admits before global completion. Positive example: sensor calibration uncertainty belongs in local credal sets. Negative example: hiding local ambiguity by picking a single precise prior can make later confidence appear unjustifiably high.

Compatible completion / natural extension. A family of finite-window lower previsions must be globally compatible before it gives a raw all-gambles process law. F2 shows that the exchangeable external readout is unconditional, while the raw crown is exactly the lower-prevision compatibility boundary. Positive example: finite-prefix forecasts can be useful before full natural extension is available. Negative example: calling those forecasts a full process law without compatibility is assumption laundering.

Completion-level extreme point. Given a fixed credal set, the world may instantiate one precise completion. This is real aleatoric freedom, but it is invisible to a truth value because $\langle s, c \rangle$ is a property of the whole credal set. Positive example: scenario simulation samples an extreme completion. Negative example: inference rules should not pretend to know which extreme completion the world selected.

Phase-transition / infinite boundary freedom. Infinite local specifications can have multiple DLR completions. F3 pins a collapse regime by Dobrushin uniqueness; F4 pins a strict-width regime by low-temperature Ising separation. Positive example: weakly coupled large sensor networks can have precise local-query envelopes. Negative example: strongly coupled spatial contagion or image domains can keep interval width because distant boundary conditions matter.

Exchangeability / de Finetti family freedom. In the IID lane, a mixing family can have a canonical external sigma-additive readout while the raw all-gambles crown depends on lower-prevision compatibility. Pointwise zero-interior mass is sufficient in some kernel cases but not necessary at the family level. Positive example: a family with one compatible conservative member can support the raw crown. Negative example: requiring every member to have zero interior mass rejects legitimate imprecise families.

Rule-level confidence transport. PLN rule formulas must decide whether confidence is combined by addition in weight space, minimum in weight space, multiplication, or a rule-specific effective evidence term. Positive example: revision adds independent evidence weights. Negative example: feeding a raw confidence through $w2c$ as if it were already a weight double-damps the result.

Provenance and route semantics. Two derivations with the same final STV may have different provenance: direct observation, pathway transfer, ab-

duction, or long inference chain. Positive example: biomedical drug support should distinguish direct disease evidence from pathway-mediated support. Negative example: merging all routes before preserving provenance makes audit and discounting impossible.

Extra structure beyond classical STV. Standard STV forgets structures such as phase, dependence witnesses, or higher-order distributional shape. The formal amplitude/phase boundary shows that such information can be invisible to the classical strength and confidence view. Positive example: interference-like composition requires extra structure. Negative example: carrying phase in an ordinary binary classifier may be needless complexity.

B Example coverage plan for PeTTa/CeTTa

These examples accompany the paper as a runnable companion package, `papers/pln-confidence-examples/` (build and run via `run_all.sh` on CeTTa(HE) and PeTTa; `DOF-INDEX.md` is the full map). Each degree of freedom falls in one of three honesty tiers: *Runnable* (a real executable canary with a positive and a negative case), *Structural* (a finite shadow of an infinite or joint fact — the Lean theorem is the real claim), and *Lean-ref* (the witness is a theorem; no standalone `.metta`). Thirteen runnable canaries (DoF 1–8, 10, 11, 13, 17, 18) are verified on both engines; the four new Lean oracles (DoF 8, 11, 14, 18) live in `Mettapedia/Logic/PLNConfidenceDoFCanary.lean` and are axiom-clean. The table below records each row’s current artifact; entries marked as planned in earlier drafts are now realized in the package.

Freedom		Current artifact	Example to keep or add
Evidence versus ratio	total	<code>Mettapedia/papers/examples/RavenAsymmetricInduction.metta</code>	Keep the raven asymmetry example; add explicit confidence readouts for 5/0, 5/95, and 50/950 under at least two κ values.
Evidence scale κ		<code>sandbox-runtime/petta-he-66ef196/tests/support/pln_portable/pln_evidence.metta</code>	Add a small table showing the same evidence count under $\kappa = 1, 10, 50$. Use curated clinical, noisy omics, and pathway-mediated labels.
Strength context	prior	Lean canaries in <code>PLNTruthTower</code>	Add a PeTTa/CeTTa count example where MLE, uniform prior, and Jeffreys prior produce different strength displays from the same sparse counts.

Freedom	Current artifact	Example to keep or add
Confidence chart freedom	<code>reserveHalfCoordinate</code> <code>not_walley_width_complement</code> in Lean	Add a runnable chart comparison: PLN odds chart versus reserve-half display, same decoded evidence, different displayed confidence.
Revision as evidence addition	<code>sandbox-runtime/petta-he-66ef196/lib/lib_pln.metta</code>	Keep <code>Truth_Revision</code> ; add a canary that revising two STVs equals adding decoded evidence weights and re-encoding.
Rule-level confidence transport	<code>Mettapedia/lean/mettapedia/Mettapedia/Logic/PeTTaLibPLNFormalAnalysis.lean</code>	Add a PeTTa regression for the raw-min double-damping bug: old formula <code>h1000</code> . weight-space formula on unit evidence.
Interval point selector	Lean generic ITV selector canary	Add a three-selector example showing lower, midpoint, and upper produce different displayed strengths on the same interval.
Width-complement range normalization	Paper convention	Add binary and dollar-utility examples: binary width uses $1 - (U - L)$; bounded utility uses $1 - (U - L)/(b - a)$.
Credal shape beyond one interval	Lean credal projection tower canaries	Add two small finite credal sets with the same query interval and different joint/correlation behavior.
Backend choice	Lean typed ITV constructors	Add side-by-side Bayesian credible interval and Walley IDM interval from the same counts, with provenance printed in the result.
Categorical query choice	Lean categorical evidence tower	Add a three-category classifier where total evidence is high but the rare category remains wide.
Feature/evidence discounting	<code>pln_evidence.metta</code> power-scaling comments	Add IDF/source-quality examples: duplicate text-mined hits discounted, curated independent hits amplified.

Freedom		Current artifact	Example to keep or add
Biomedical route provenance		<code>sandbox-runtime/petta-he-66ef196/corpus/cetta_tests/test_bio_wmpln_pathway_route_regression.metta</code>	Keep direct-versus-pathway route tests; add route-specific κ or discounting before supported-drug revision.
Exchangeable de Finetti F2		Lean F2 crown	Add a finite-prefix coin/patient-response stream: external read-out always exists; raw all-gambles crown marked as requiring compatibility. Include the counterexample to pointwise zero-interior necessity as a symbolic canary.
Dobrushin collapse	F3	Lean F3 crown	Add a weakly coupled sensor-grid story: all finite-query completions agree under the Dobrushin budget, so the interval collapses.
Ising width	F4 strict	Lean F4 crown	Add a spatial contagion or image-segmentation story: low-temperature coupling and boundary conditions keep the origin query interval nontrivial.
Completion-level extreme point		Credal-set semantics	Add a scenario-sampling example where two precise completions are sampled from the same credal set but the displayed STV remains the envelope summary.
Extra structure	hidden	Amplitude/phase Lean boundary	Add an optional example with two hidden-phase states that share classical STV but differ under a composition rule that sees phase.

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